

Yichao Cai

yichao.cai@adelaide.edu.au · +61 478 927 202 · yichaocai.com · GitHub · Google Scholar · LinkedIn

Research Statement

I work on understanding when modern learning objectives recover latent structure beyond predictive performance, using tools from identifiability theory, latent-variable modeling, and representation geometry. In particular, I study the equivalence classes of representations induced by learning objectives, and how cross-modal supervision shapes the geometry of vision-language models. I am also interested in how these learned representations relate to human-interpretable concepts.

Education

Ph.D. in Computer Science

Adelaide University (formerly The University of Adelaide)

Supervisor: Prof. Javen Qinfeng Shi

Sep 2023 – Present

M.Sc. in Instrument Science and Technology

Wuhan University of Technology

Supervisor: Prof. Xiao Zhou

Sep 2016 – Jun 2019

B.Eng. in Measurement & Control Technology and Instrument

Wuhan University of Technology

Sep 2012 – Jun 2016

Experience

PhD Student Researcher

Australian Institute for Machine Learning (AIML), Adelaide University

Sep 2023 – Present

- Developed a theoretical framework for the geometry of InfoNCE-based contrastive learning, characterising alignment potentials, entropic dispersion forces, and the emergence of modality gap (ICMLR 2026).
- Studied cross-modal misalignment as a beneficial inductive bias in multimodal contrastive learning; established identifiability conditions and controlled benchmarks (NeurIPS 2025 Spotlight).
- Developed CLAP, a contrastive representation learning framework for content–style disentanglement via augmented language prompts (ECCV 2024).

AI Engineer

Tellhow Software

Jun 2020 – Aug 2022

- Built inspection and safety-analytics solutions for power-industry equipment using computer vision and automated analysis workflows; delivered end-to-end features from prototyping to production deployment.

Software Engineer

Huawei Technologies

Jul 2019 – Apr 2020

- Developed communication service components; implemented production features and system integrations.

Visiting Student Researcher

California PATH, UC Berkeley

May 2018 – Sep 2018

- Conducted research on visual perception algorithms for autonomous driving.

Teaching

Teaching Assistant - Neural Networks and Deep Learning (ARTI X300), Adelaide University

Semester 1 2026

Guest Lecturer and Head Tutor - Statistical Machine Learning (COMP SCI 3314), Adelaide University

Semester 2 2025

Teaching Assistant - Using Machine Learning Tools (COMP SCI 7317), Adelaide University

Trimester 2 2025

Teaching Assistant - Concepts in AI and ML (COMP SCI 7327), Adelaide University

Semester 1 2025

Publications

- [1] **Yichao Cai**, Zhen Zhang, Yuhang Liu, Javen Qinfeng Shi. **The Geometric Mechanics of Contrastive Representation Learning: Alignment Potentials, Entropic Dispersion, and Cross-Modal Divergence.** *International Conference on Machine Learning (ICML)*, 2026.
- [2] Wenkang Jiang, Yuhang Liu, **Yichao Cai**, Erdun Gao, Jiayi Dong, Ehsan Abbasnejad, Lina Yao, Javen Qinfeng Shi. **What Makes a Good Representation for Single-Cell Perturbation Prediction?** *International Conference on*

Machine Learning (ICML), 2026.

- [3] Jiaqing Chen, Zidu Yin, **Yichao Cai**, Yuhang Liu, Zhen Zhang, Dong Gong, Javen Qinfeng Shi. **Boundary Embedding Shaping with Adaptive Contrastive Learning for Graph Structural Disentanglement.** *International Conference on Machine Learning (ICML), 2026.*
- [4] Yuhang Liu, Dong Gong, **Yichao Cai**, Erdun Gao, Zhen Zhang, Biwei Huang, Mingming Gong, Anton van den Hengel, Javen Qinfeng Shi. **I Predict Therefore I Am: Is Next Token Prediction Enough to Learn Human-Interpretable Concepts from Data?** *International Conference on Learning Representations (ICLR), 2026.*
- [5] **Yichao Cai***, Yuhang Liu*, Erdun Gao, Tianjiao Jiang, **Zhen Zhang**, **Anton van den Hengel**, **Javen Qinfeng Shi**. **On the Value of Cross-Modal Misalignment in Multimodal Representation Learning.** *Advances in Neural Information Processing Systems (NeurIPS), 2025 (Spotlight).* *Equal contribution.
- [6] **Yichao Cai**, Yuhang Liu, Zhen Zhang, Javen Qinfeng Shi. **CLAP: Isolating Content from Style Through Contrastive Learning with Augmented Prompts.** *European Conference on Computer Vision (ECCV), 2024.*

Academic Service

Conference Reviewers

- Conference on Neural Information Processing Systems (NeurIPS) 2026
- International Conference on Machine Learning (ICML) 2026, Silver Reviewer Award
- International Conference on Learning Representations (ICLR) 2026

Journal Reviewers

- Transactions on Machine Learning Research (TMLR)

Honors and Awards

NeurIPS 2025 Scholar Award, Neural Information Processing Foundation	Oct 2025
University of Adelaide Research Scholarships	Sep 2023
Award for Outstanding Graduates, Wuhan University of Technology	Jun 2019

Miscellaneous

Reading, guitar, and nature walks.